MIPT & YSDA · AUTUMN 2026

# Deep Generative Models

## Lecture 9

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# Recap of Previous Lecture

## ELBO for Gaussian Diffusion Model

$$\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) = \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \log p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1) - D_{\mathrm{KL}}\big(q(\mathbf{x}_T|\mathbf{x}_0)\|p(\mathbf{x}_T)\big) - \sum_{t=2}^{T} \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\mathrm{KL}}\big(q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)\|p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)\big)}_{\mathcal{L}_t}$$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}|\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t,\mathbf{x}_0),\tilde{\beta}_t\mathbf{I}),$$
$$p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}\big(\mathbf{x}_{t-1}|\boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t),\boldsymbol{\sigma}^2_{\boldsymbol{\theta},t}(\mathbf{x}_t)\big)$$

It is assumed that $\boldsymbol{\sigma}^2_{\boldsymbol{\theta},t}(\mathbf{x}_t) = \tilde{\beta}_t\mathbf{I}$.

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}\left[\frac{1}{2\tilde{\beta}_t}\left\|\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t,\mathbf{x}_0) - \boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t)\right\|^2\right]$$

# Recap of Previous Lecture

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[ \frac{1}{2\tilde{\beta}_t} \left\| \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t) \right\|^2 \right]$$

## Reparametrization

$$\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \boldsymbol{\epsilon}$$

$$\boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t) = \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \boldsymbol{\epsilon}_{\boldsymbol{\theta},t}(\mathbf{x}_t)$$

# Recap of Previous Lecture

$$\mathcal{L}_t = \mathbb{E}_{\boldsymbol{\epsilon}\sim\mathcal{N}(0,\mathbf{I})}\left[\frac{(1-\alpha_t)^2}{2\tilde{\beta}_t\alpha_t(1-\bar{\alpha}_t)}\left\|\boldsymbol{\epsilon}-\boldsymbol{\epsilon}_{\boldsymbol{\theta},t}\left(\sqrt{\bar{\alpha}_t}\mathbf{x}_0+\sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}\right)\right\|^2\right]$$

Predict the added noise. Dropping the time-dependent ELBO coefficient changes the relative weights of the training levels.

## Simplified Objective

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t\sim U\{1,T\}}\mathbb{E}_{\boldsymbol{\epsilon}\sim\mathcal{N}(0,\mathbf{I})}\left\|\boldsymbol{\epsilon}-\boldsymbol{\epsilon}_{\boldsymbol{\theta},t}\left(\sqrt{\bar{\alpha}_t}\cdot\mathbf{x}_0+\sqrt{1-\bar{\alpha}_t}\cdot\boldsymbol{\epsilon}\right)\right\|^2$$

# Recap of Previous Lecture

## Training

1. Sample $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$, $t \sim U\{1, T\}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$.
2. Compute noisy image $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \boldsymbol{\epsilon}$.
3. Compute loss $\mathcal{L}_{\text{simple}} = \|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_{\boldsymbol{\theta}, t}(\mathbf{x}_t)\|^2$.

## Sampling (Ancestral)

1. Sample $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$.
2. Compute the mean of $p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\boldsymbol{\mu}_{\boldsymbol{\theta}, t}(\mathbf{x}_t), \tilde{\beta}_t \cdot \mathbf{I})$:

$$\boldsymbol{\mu}_{\boldsymbol{\theta}, t}(\mathbf{x}_t) = \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \boldsymbol{\epsilon}_{\boldsymbol{\theta}, t}(\mathbf{x}_t).$$

3. Denoise $\mathbf{x}_{t-1} = \boldsymbol{\mu}_{\boldsymbol{\theta}, t}(\mathbf{x}_t) + \sqrt{\tilde{\beta}_t} \cdot \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$.

# Recap of Previous Lecture

Write the conditional score error as

$$\ell_{\boldsymbol{\theta}}(t) = \mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left\| \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}_0) \right\|^2 .$$

**Same regression form, different weights**

$$\mathcal{L} = \mathbb{E}_{t \sim U\{1,T\}} \left[ w_t \, \ell_{\boldsymbol{\theta}}(t) \right] .$$

| Objective | Gaussian perturbation | Score-loss weight $w_t$ |
|---|---|---|
| DDPM simple loss | $\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}$ | $1 - \bar{\alpha}_t$ |
| NCSN | $\mathbf{x}_0 + \sigma_t \boldsymbol{\epsilon}$ | $\sigma_t^2$ |

The DDPM ELBO uses different weights. Replacing them changes the objective.

Ho J., et al. Denoising Diffusion Probabilistic Models, 2020
Song Y., Ermon S. Generative Modeling by Estimating Gradients of the Data Distribution, 2019

# Recap of Previous Lecture

## Loss Weighting and Time Sampling

For fixed $w_t$ and expected per-step losses $\ell_{\boldsymbol{\theta}}(t)$:

$$\mathcal{L} = \frac{1}{T}\sum_{t=1}^{T} w_t \ell_{\boldsymbol{\theta}}(t) = \mathbb{E}_{t\sim r}\left[\frac{w_t}{T r_t}\ell_{\boldsymbol{\theta}}(t)\right], \qquad r_t > 0.$$

- Loss weights determine which noise levels matter in the objective.
- With the correction $1/(Tr_t)$, changing the sampling probabilities preserves the objective.
- Without correction, changing the probabilities changes the relative weights.
- Training-time sampling and the generation grid are separate choices.

# Recap of Previous Lecture

## Summary

- Gaussian perturbations:
  - DDPM: $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \boldsymbol{\epsilon}$;
  - NCSN: $\mathbf{x}_t = \mathbf{x}_0 + \sigma_t \cdot \boldsymbol{\epsilon}$.
  - Rescaling the DDPM state by $1/\sqrt{\bar{\alpha}_t}$ gives the NCSN noise form.
- Both use denoising score matching, with objective-specific noise-level weights.
- The sampling procedures differ:
  - Ancestral sampling in DDPM;
  - Annealed Langevin dynamics for NCSN;
  - Hybrid approaches that combine both updates are possible.

Kingma D. et al. Variational Diffusion Models, 2021
Song Y. et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

# Recap of Previous Lecture

## Sampling (DDIM)

1. Sample $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$; choose a grid $T = t_K > \cdots > t_0 = 0$.
2. For $k = K, \ldots, 1$, set $t = t_k$, $s = t_{k-1}$ and compute:

$$\hat{\boldsymbol{\epsilon}} = \boldsymbol{\epsilon}_{\boldsymbol{\theta},t}(\mathbf{x}_t) \qquad \text{predict noise,}$$

$$\hat{\mathbf{x}}_0 = \frac{\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\,\hat{\boldsymbol{\epsilon}}}{\sqrt{\bar{\alpha}_t}} \qquad \text{estimate clean data,}$$

$$\mathbf{x}_s = \sqrt{\bar{\alpha}_s}\,\hat{\mathbf{x}}_0 + \sqrt{1-\bar{\alpha}_s}\,\hat{\boldsymbol{\epsilon}} \qquad \text{update state.}$$

3. Return $\mathbf{x}_0$.

**Deterministic limit:** fresh transition noise is zero ($\sigma_{t\to s} = 0$); randomness comes from $\mathbf{x}_T$.

**Fewer steps:** choose $K \ll T$; $K$ steps require $K$ network evaluations and no retraining.

# Recap of Previous Lecture

## DDPM Sampling

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1-\beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1-\beta_t}} \cdot \nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t) + \sigma_t \cdot \boldsymbol{\epsilon}$$

## Guided Generation

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1-\beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1-\beta_t}} \cdot \nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t|\mathbf{y}) + \sigma_t \cdot \boldsymbol{\epsilon}$$

## Guided Generation

$$\begin{aligned}\nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t|\mathbf{y}) &= \nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t) \\ &= \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)\end{aligned}$$

Here, $p(\mathbf{y}|\mathbf{x}_t)$ denotes a classifier operating on noisy samples (which must be trained separately).

# Recap of Previous Lecture

## Guided Score Function

$$\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y}) = \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$$
$$\mathbf{s}^{\gamma}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y}) = \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$$

**Note:** Increasing $\gamma$ sharpens $p(\mathbf{y}|\mathbf{x}_t)$, increasing the contrast

$$\hat{p}(\mathbf{y}|\mathbf{x}_t) \propto p(\mathbf{y}|\mathbf{x}_t)^{\gamma}.$$

Very large $\gamma$ can reduce diversity and introduce artifacts.

## Guidance Interval

Apply extra guidance only at intermediate noise levels. Outside the interval, $\gamma_t = 1$: **conditioning remains**.

Dhariwal P., Nichol A. Diffusion Models Beat GANs on Image Synthesis, 2021
Kynkäänniemi T. et al. Applying Guidance in a Limited Interval Improves Sample and Distribution Quality in Diffusion Models, 2024 · Sadat S. et al. Eliminating Oversaturation and Artifacts of High Guidance Scales in Diffusion Models, 2025

# Recap of Previous Lecture

## Training

1. Train the DDPM as before.
2. Train an additional classifier $p(\mathbf{y}|\mathbf{x}_t)$ on noisy data (time-dependent).

## Sampling (Guided)

1. Sample $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$.
2. Denoise with the scaled guided score function:

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1-\beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1-\beta_t}} \cdot \mathbf{s}^{\gamma}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y}) + \sigma_t \cdot \boldsymbol{\epsilon}.$$

# Recap of Previous Lecture

The previous method requires an additional classifier $p(\mathbf{y}|\mathbf{x}_t)$ trained on noisy data. Let's try to avoid this requirement.

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t) = \nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t|\mathbf{y}) - \nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t)$$

$$\begin{aligned}
\nabla^{\gamma}_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t|\mathbf{y}) &= \nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t) \\
&= \nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t) + \gamma \cdot \big(\nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t|\mathbf{y}) - \nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t)\big) \\
&= (1-\gamma) \cdot \nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t|\mathbf{y})
\end{aligned}$$

Ho J., Salimans T. Classifier-Free Diffusion Guidance, 2022
Holderrieth P., Erives E. An Introduction to Flow Matching and Diffusion Models, 2025

# Recap of Previous Lecture

## Scaled Guided Score Function

$$\mathbf{s}^{\gamma}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y}) = (1-\gamma) \cdot \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) + \gamma \cdot \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y})$$

## CFG

1. Introduce the "absence of conditioning" label $\mathbf{y} = \varnothing$.
2. Identify the unguided score function $\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) = \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \varnothing)$.
3. Train a single model $\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y})$ on **supervised** data, dropping the label $\mathbf{y}$ with some fixed probability (simulating $\mathbf{y} = \varnothing$).
4. At inference, evaluate the model twice to obtain $\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \varnothing)$ and $\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y})$.

**Guidance distillation:** train a student to match the fixed teacher's guided prediction:

$$\mathbf{s}_{\boldsymbol{\phi},t}(\mathbf{x}_t, \mathbf{y}, \gamma) \approx (1-\gamma)\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \varnothing) + \gamma \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y}).$$

Evaluate the student **once per step**; keep the sampler and step count.

Ho J., Salimans T. Classifier-Free Diffusion Guidance, 2022
Holderrieth P., Erives E. An Introduction to Flow Matching and Diffusion Models, 2025
Meng C. et al. On Distillation of Guided Diffusion Models, 2023

# Outline

01 Continuous-Time Normalizing Flows (CNF)

02 Continuity Equation for CNF Log-Likelihood

03 SDE Basics

# Outline

01 **Continuous-Time Normalizing Flows (CNF)**

02 Continuity Equation for CNF Log-Likelihood

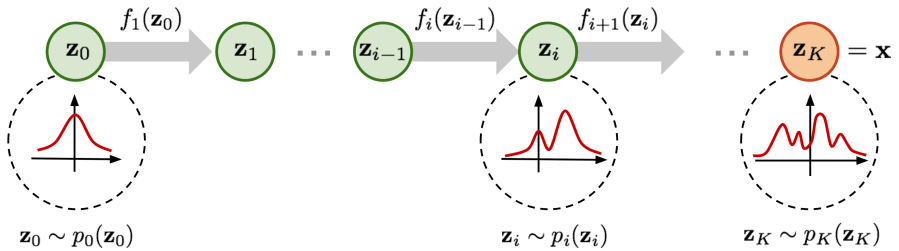
03 SDE Basics

# Discrete-Time Normalizing Flows

## Change of Variable Theorem (CoV)

Let $\mathbf{x}$ be a random variable with density $p(\mathbf{x})$, and let $\mathbf{f} : \mathbb{R}^m \to \mathbb{R}^m$ be a differentiable and **invertible** transformation. If $\mathbf{z} = \mathbf{f}(\mathbf{x})$, $\mathbf{x} = \mathbf{f}^{-1}(\mathbf{z}) = \mathbf{g}(\mathbf{z})$, then

$$
\begin{aligned}
p(\mathbf{x}) &= p(\mathbf{z}) |\det(\mathbf{J}_\mathbf{f})| = p(\mathbf{z}) \left| \det \left( \frac{\partial \mathbf{z}}{\partial \mathbf{x}} \right) \right| \\
&= p(\mathbf{f}(\mathbf{x})) \left| \det \left( \frac{\partial \mathbf{f}(\mathbf{x})}{\partial \mathbf{x}} \right) \right|
\end{aligned}
$$




$$
\log p_{\boldsymbol{\theta}}(\mathbf{x}) = \log p(\mathbf{f}_K \circ \cdots \circ \mathbf{f}_1(\mathbf{x})) + \sum_{k=1}^{K} \log \left| \det \left( \frac{\partial \mathbf{f}_k}{\partial \mathbf{f}_{k-1}} \right) \right| .
$$

# Towards Continuous-Time Normalizing Flows

Up to this point, we have considered discrete-time normalizing flows:

$$\mathbf{x}_{t+1} = \mathbf{f}_{\boldsymbol{\theta}}(\mathbf{x}_t, t); \quad \log p(\mathbf{x}_{t+1}) = \log p(\mathbf{x}_t) - \log \left| \det \frac{\partial \mathbf{f}_{\boldsymbol{\theta}}(\mathbf{x}_t)}{\partial \mathbf{x}_t} \right|.$$

## Residual Flows

Here $\mathbf{f}_{\boldsymbol{\theta}}$ is the NF **bijection**, $\mathbf{v}_{\boldsymbol{\theta}}$ is the **velocity** (vector field):

$$\mathbf{x}_{t+1} = \mathbf{f}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) = \mathbf{x}_t + \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)$$

Given $\mathbf{y} = \mathbf{f}_{\boldsymbol{\theta}}(\mathbf{x}, t)$, recover $\mathbf{x}$ by **fixed-point iteration**:

$$\mathbf{x} = \mathbf{y} - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t), \qquad \mathbf{x}^{(k+1)} = \mathbf{y} - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}^{(k)}, t).$$

If $\mathbf{v}_{\boldsymbol{\theta}}$ has Lipschitz constant $L < 1$ in $\mathbf{x}$, the iteration is contractive. **Banach's theorem** guarantees a unique solution for every $\mathbf{y}$, so $\mathbf{f}_{\boldsymbol{\theta}}$ is invertible.

Behrmann J. et al. Invertible Residual Networks, 2019
Chen R. T. Q. et al. Residual Flows for Invertible Generative Modeling, 2019

# Towards Continuous-Time Normalizing Flows

A residual step $\mathbf{x}(t+h) = \mathbf{x}(t) + h\,\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)$ is invertible if $hL < 1$:

$$\frac{\mathbf{x}(t+h) - \mathbf{x}(t)}{h} = \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)$$

Taking the limit $h \to 0$, we obtain continuous-time dynamics.

## Continuous-Time Dynamics

Consider an Ordinary Differential Equation (ODE):

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t); \quad \text{with initial condition } \mathbf{x}(t_0) = \mathbf{x}_0.$$

$$\mathbf{x}(t_1) = \int_{t_0}^{t_1} \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)\,dt + \mathbf{x}_0$$

The **velocity field** $\mathbf{v}_{\boldsymbol{\theta}} : \mathbb{R}^m \times [t_0, t_1] \to \mathbb{R}^m$ need not be contractive or invertible. Under suitable regularity conditions, **uniqueness of ODE solutions guarantees an invertible flow**.

# ODEs, Flows and Numerical Solutions

## Ordinary Differential Equation

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t), \qquad \mathbf{x}(t_0) = \mathbf{x}_0.$$

## Flow

The **flow** $\boldsymbol{\psi} : \mathbb{R}^m \times [t_0, t_1] \to \mathbb{R}^m$ maps the initial point to its position at time $t$: $\mathbf{x}(t) = \boldsymbol{\psi}_t(\mathbf{x}_0)$.

$$\frac{d\boldsymbol{\psi}_t(\mathbf{x}_0)}{dt} = \mathbf{v}_{\boldsymbol{\theta}}(\boldsymbol{\psi}_t(\mathbf{x}_0), t), \qquad \boldsymbol{\psi}_{t_0}(\mathbf{x}_0) = \mathbf{x}_0.$$

## Numerical Solution

$$\boldsymbol{\psi}_t(\mathbf{x}_0) = \mathbf{x}_0 + \int_{t_0}^{t} \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(s), s)\, ds \approx \texttt{ODESolve}_v(\mathbf{x}_0, \boldsymbol{\theta}, t_0, t).$$

A numerical solver approximates the flow by a sequence of **finite steps**.

# Numerical Solution of ODEs: Euler

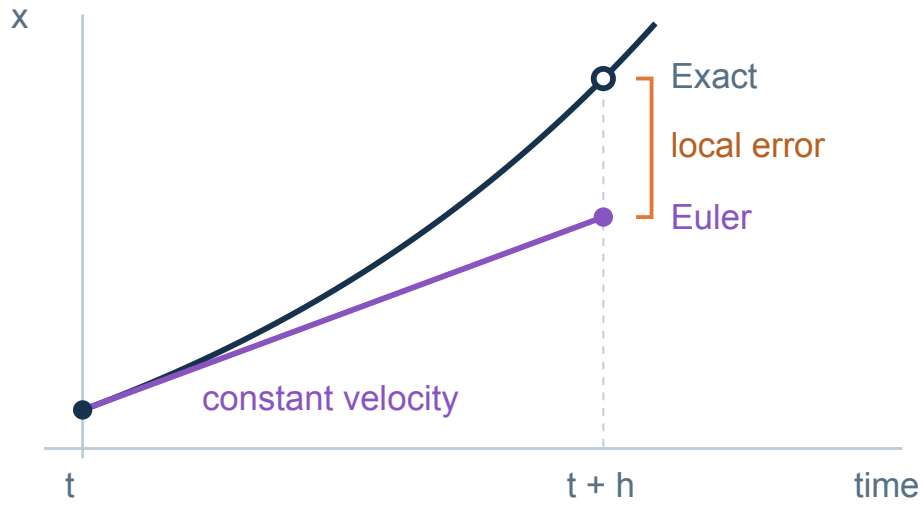
## Euler Update Step

$$\frac{\mathbf{x}(t+h) - \mathbf{x}(t)}{h} = \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)$$

$$\mathbf{x}(t+h) = \mathbf{x}(t) + h \cdot \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)$$

Freeze the current velocity over the step: **one field evaluation**.

**One step: follow the tangent**




## One-Step Error

For a smooth field, Taylor expansion from the **exact starting point** gives

$$\mathbf{x}(t+h) = \mathbf{x}(t) + h\,\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t) + \mathcal{O}(h^2).$$

Euler drops the higher-order terms: the **local error is** $\mathcal{O}(h^2)$.

Euler method; original one-step illustration
Driscoll T. A., Braun R. J. Fundamentals of Numerical Computation: Runge–Kutta methods

# Numerical Solution of ODEs: Euler and Heun

Same initial point and step size: 8 steps on [0, 3], h = 0.375.

## Euler

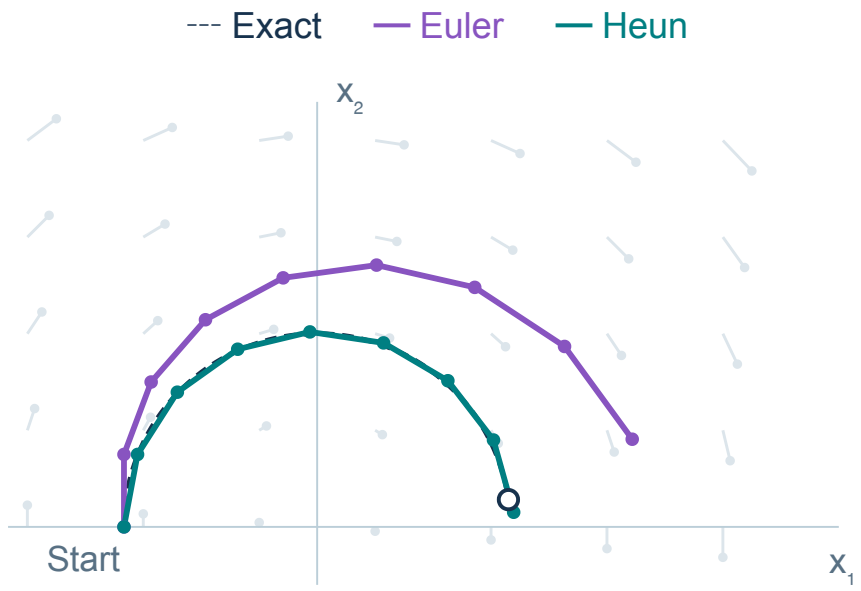
$$\mathbf{x}(t+h) = \mathbf{x}(t) + h\,\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)$$

## Heun: predict the endpoint

$$\mathbf{x}'(t+h) = \mathbf{x}(t) + h\,\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)$$

## Heun: average the two velocities

$$\mathbf{x}(t+h) = \mathbf{x}(t) + \frac{h}{2}\big(\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t) + \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}'(t+h), t+h)\big)$$




Heun averages two velocities.
Twice the field evaluations per step.

Error 0.713 Error 0.071

Step 8 / 8 · Field evaluations: Euler 8, Heun 16.

Example: $\mathbf{v}(\mathbf{x}) = (x_2, -x_1), \quad \mathbf{x}(0) = (-1, 0), \quad t \in [0, 3].$

# ODE Solvers: Accuracy and Cost

For a sufficiently smooth velocity field on a fixed interval $[0, T]$:

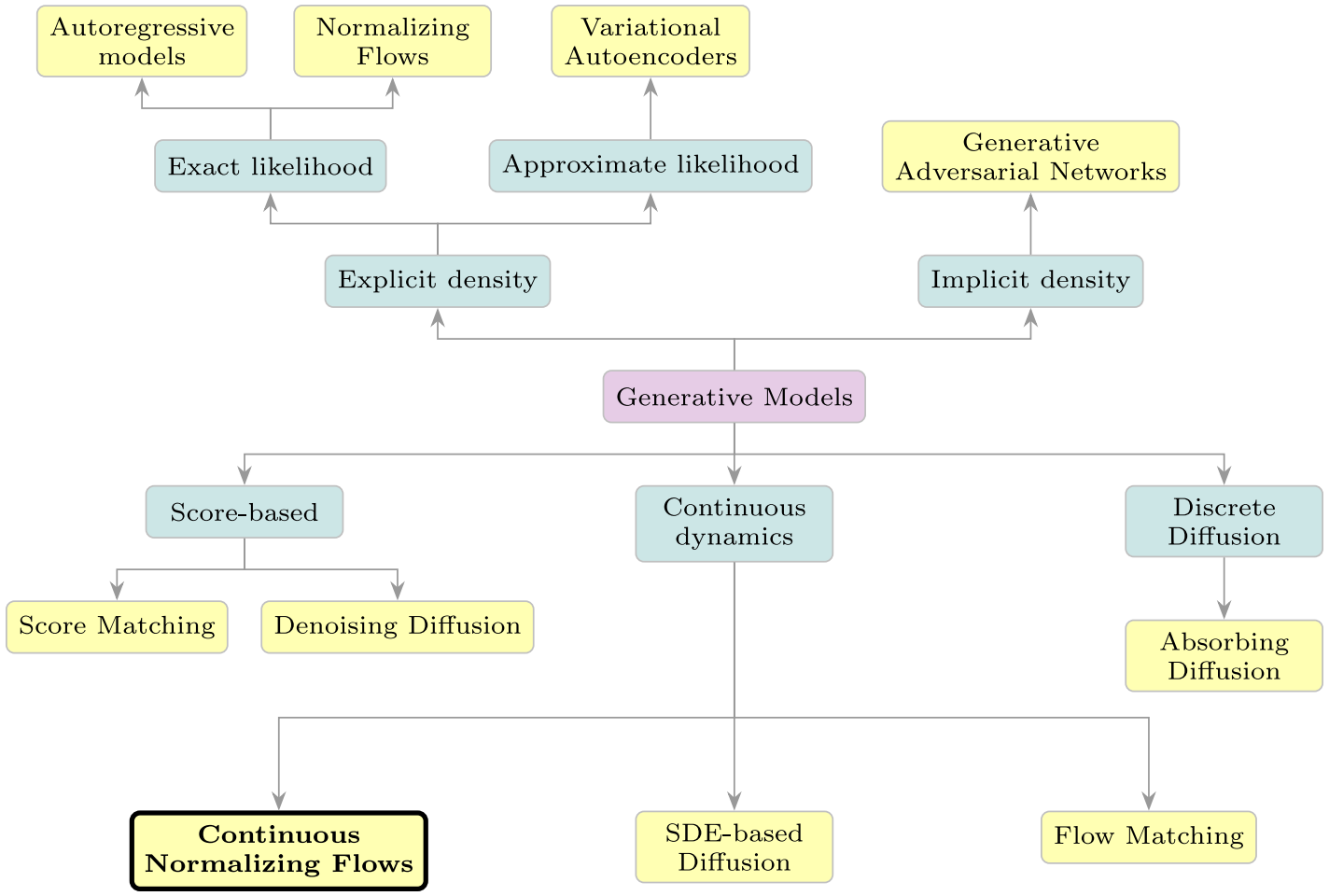
$$\|\widehat{\mathbf{x}}_N - \mathbf{x}(T)\| = \mathcal{O}(h^p), \qquad h = T/N.$$

This is the **global error** after $N$ steps.

| Method | Global error | Velocity evaluations / step |
|---|---|---|
| Euler | $\mathcal{O}(h)$ | 1 |
| Heun | $\mathcal{O}(h^2)$ | 2 |
| RK4 | $\mathcal{O}(h^4)$ | 4 |

- **One-step error** from an exact starting point is $\mathcal{O}(h^{p+1})$.
- For small $h$, halving the step roughly reduces global error by **2 / 4 / 16**, but doubles the number of steps.
- Compare computational cost by **NFE**: the number of velocity-network evaluations.

# Generative Models Taxonomy



# Continuous-Time Normalizing Flows: Neural ODE

## Neural ODE

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t); \quad \text{with initial condition } \mathbf{x}(t_0) = \mathbf{x}_0$$

## Euler `ODESolve`

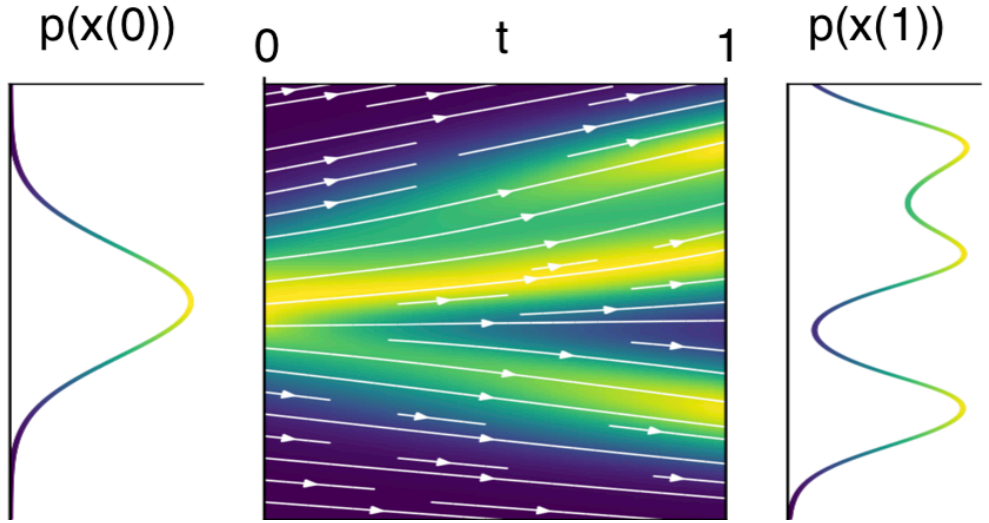
$$\mathbf{x}(t + h) = \mathbf{x}(t) + h \cdot \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)$$

- Consider $[t_0, t_1] = [0, 1]$ for simplicity.
- If $\mathbf{x}(0)$ is a random variable with density $p_0(\mathbf{x})$,
- Then, for any $t$, $\mathbf{x}(t)$ is a random variable with density $p_t(\mathbf{x})$.

# Continuous-Time Normalizing Flows: Intuition

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t); \quad \text{with initial condition } \mathbf{x}(t_0) = \mathbf{x}_0$$

- $p_t(\mathbf{x}) = p(\mathbf{x}, t)$ describes the **probability path** interpolating between $p_0(\mathbf{x})$ and $p_1(\mathbf{x})$.
- What is the difference between $p_t(\mathbf{x}(t))$ and $p_t(\mathbf{x})$?



# Continuous-Time Normalizing Flows: Reversibility

**Theorem (Picard)**

If $\mathbf{v}$ is continuously differentiable with a bounded derivative in $\mathbf{x}$ and continuous in $t$, then the ODE has a **unique solution** given by a flow $\boldsymbol{\psi}_t$.

This guarantees the ODE is **uniquely reversible**.

$$\boldsymbol{\psi}_1(\mathbf{x}_0) = \mathbf{x}_0 + \int_0^1 \mathbf{v}_{\boldsymbol{\theta}}(\boldsymbol{\psi}_t(\mathbf{x}_0), t)\, dt$$

$$\mathbf{x}(1) = \mathbf{x}(0) + \int_0^1 \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)\, dt$$

$$\mathbf{x}(0) = \mathbf{x}(1) + \int_1^0 \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)\, dt$$

**Note:** Unlike discrete-time flows, $\mathbf{v}$ need not be invertible (uniqueness ensures bijection).

How can we compute $p_t(\mathbf{x})$ at arbitrary $t$?

# Outline

01 Continuous-Time Normalizing Flows (CNF)

**02 Continuity Equation for CNF Log-Likelihood**

03 SDE Basics

# Continuous-Time NF

### Theorem (Continuity Equation)

If $\mathbf{v}$ is continuously differentiable in $\mathbf{x}$ (with bounded derivative) and continuous in $t$, and $p_t(\mathbf{x}) > 0$, then

$$\frac{d \log p_t(\mathbf{x}(t))}{dt} = -\operatorname{tr}\left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right)$$

Integrating gives the density **along the trajectory**:

### Solution

$$\log p_1(\mathbf{x}(1)) = \log p_0(\mathbf{x}(0)) - \int_0^1 \operatorname{tr}\left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right) dt.$$

### Hutchinson's trace estimator:

$$\operatorname{tr}(\mathbf{J}) = \mathbb{E}_{\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})}[\boldsymbol{\epsilon}^T \mathbf{J} \boldsymbol{\epsilon}], \qquad \mathbf{J} = \frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}.$$

- One noise vector gives an **unbiased trace estimate** via autodiff, without forming the full Jacobian.
- **Exact identity; approximate computation** using an ODE solver and a stochastic trace estimate.

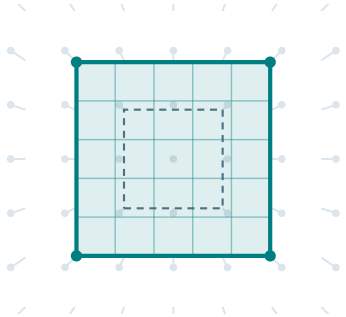
Chen R. T. Q. et al. Neural Ordinary Differential Equations, 2018
Grathwohl W. et al. FFJORD: Free-form Continuous Dynamics for Scalable Reversible Generative Models, 2018

# Divergence: Where Does the Density Go?

**Lecture 2: area change** $\dfrac{A_t}{A_0} = |\det \mathbf{J}_{\psi_t}|, \qquad \mathbf{J}_{\psi_t} = \dfrac{\partial \psi_t}{\partial \mathbf{x}_0}$

**CNF: log-area rate** $\dfrac{d}{dt} \log |\det \mathbf{J}_{\psi_t}| = \operatorname{div} \mathbf{v} = \operatorname{tr}\left(\dfrac{\partial \mathbf{v}}{\partial \mathbf{x}}\right)$

**Expansion · $\mathbf{v} = 0.45\mathbf{x}$**



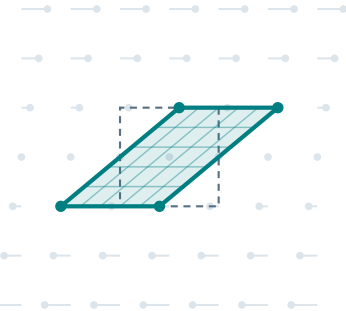
| | |
|---|---|
| Divergence | **0.90** |
| Area | **3.857** |
| Density inside the patch | **0.259** |
| Area × density | **1.000** |

**Contraction · $\mathbf{v} = -0.45\mathbf{x}$**



| | |
|---|---|
| Divergence | **-0.90** |
| Area | **0.259** |
| Density inside the patch | **3.857** |
| Area × density | **1.000** |

**Shear · $\mathbf{v} = (0.8x_2, 0)$**



| | |
|---|---|
| Divergence | **0.00** |
| Area | **1.000** |
| Density inside the patch | **1.000** |
| Area × density | **1.000** |

Mass is conserved: $\dfrac{d \log p_t(\mathbf{x}(t))}{dt} = -\dfrac{d \log A_t}{dt} = -\operatorname{div} \mathbf{v}.$

t = 1.5. Dashed: initial unit square; solid: moving patch. Area scaling is local in general; exact here (linear flows).

# CNF: Training and Sampling

Fix $p_0 = \mathcal{N}(0, \mathbf{I})$ and learn $\mathbf{v}_{\boldsymbol{\theta}}$; the endpoint density is $p_1 = p_{\boldsymbol{\theta}}$.

## Training

1. Sample $\mathbf{x} \sim p_{\text{data}}$. Solve the ODE **backward**, from $\mathbf{x}(1) = \mathbf{x}$ to $\mathbf{x}(0)$.
2. Compute the negative log-likelihood along this trajectory:

$$\mathcal{L} = -\log p_{\boldsymbol{\theta}}(\mathbf{x}) = -\log p_0(\mathbf{x}(0)) + \int_0^1 \operatorname{tr}\left(\frac{\partial \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right) dt.$$

3. Update $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} \mathcal{L}$; differentiate through the ODE, e.g., via the **adjoint method**.

## Sampling

1. Sample $\mathbf{x}(0) \sim p_0$.
2. Solve the ODE **forward** and return the endpoint:

$$\mathbf{x}(1) = \texttt{ODESolve}_v(\mathbf{x}(0), \boldsymbol{\theta}, t_0 = 0, t_1 = 1) \sim p_{\boldsymbol{\theta}}.$$

Chen R. T. Q. et al. Neural Ordinary Differential Equations, 2018
Grathwohl W. et al. FFJORD: Free-form Continuous Dynamics for Scalable Reversible Generative Models, 2018

# Outline

01 Continuous-Time Normalizing Flows (CNF)

02 Continuity Equation for CNF Log-Likelihood

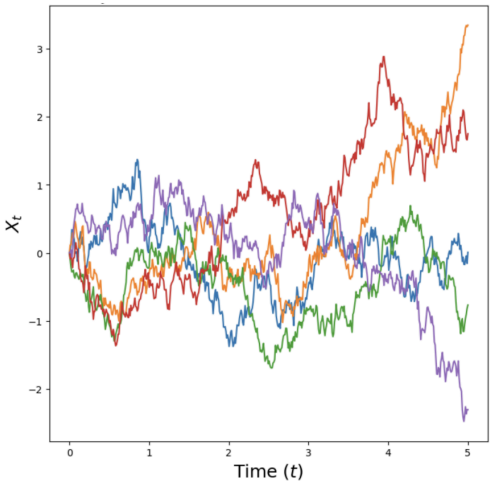
**03 SDE Basics**

# Stochastic Differential Equation (SDE)

## Wiener Process

$\mathbf{w}(t)$ is the standard Wiener process (Brownian motion), defined by:

1. $\mathbf{w}(0) = 0$ (almost surely);
2. $\mathbf{w}(t)$ has independent increments;
3. $\mathbf{w}(t)$ trajectories are continuous;
4. $\mathbf{w}(t) - \mathbf{w}(s) \sim \mathcal{N}(0, (t-s)\mathbf{I})$ for $t > s$;




$$d\mathbf{w} = \mathbf{w}(t + dt) - \mathbf{w}(t) = \mathcal{N}(0, \mathbf{I} \cdot dt) = \boldsymbol{\epsilon} \cdot \sqrt{dt}, \quad \text{where } \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}).$$

# Stochastic Differential Equation (SDE)

$$\frac{d\mathbf{x}}{dt} = \mathbf{v}(\mathbf{x}, t) \quad \Rightarrow \quad d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)\, dt$$

Let's define a stochastic process $\mathbf{x}(t)$ with initial condition $\mathbf{x}(0) \sim p_0(\mathbf{x}) = p_{\text{data}}(\mathbf{x})$:

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)\, dt + g(t)\, d\mathbf{w}$$

- $\mathbf{f}(\mathbf{x}, t) : \mathbb{R}^m \times [0, 1] \to \mathbb{R}^m$ is the **drift** term (vector field).
- $g(t) : \mathbb{R} \to \mathbb{R}$ is the **diffusion** term (if $g(t) = 0$, we recover the standard ODE).
- $\mathbf{w}(t)$ is the standard Wiener process ($d\mathbf{w} = \boldsymbol{\epsilon} \cdot \sqrt{dt}$).
- We do not have the flow $\boldsymbol{\psi}_t(\mathbf{x}_0)$ notion anymore, since trajectories are stochastic.

# Itô Integration Basics

The SDE should be understood in its **integral form** (Itô sense):

$$\mathbf{x}(t) = \mathbf{x}(0) + \int_0^t \mathbf{f}(\mathbf{x}(s), s)\, ds + \int_0^t g(s)\, d\mathbf{w}(s)$$

- The first integral is a classical (Riemann) integral.
- The second integral is an **Itô stochastic integral**.

$$\int_0^t g(s)\, d\mathbf{w}(s) = \lim_{N\to\infty} \sum_{i=0}^{N-1} g(t_i)\big(\mathbf{w}(t_{i+1}) - \mathbf{w}(t_i)\big)$$

- Brownian paths are continuous but a.s. **nowhere differentiable** $\Rightarrow$ Fundamental Theorem of Calculus fails.
- **Itô's lemma** is the stochastic counterpart of the chain rule; we will not need its explicit form here.
- Expressions $d\mathbf{x}$, $dt$, $d\mathbf{w}$ are **formal shorthand** for infinitesimal increments.

# Stochastic Differential Equation (SDE)

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)\, dt + g(t)\, d\mathbf{w}$$

**Theorem**

If $\mathbf{f}$ is continuously differentiable with a bounded derivative in $\mathbf{x}$ and continuous in $t$ and $g(t)$ is continuous then the SDE has the solution given by unique process $\mathbf{x}(t)$.

- Unlike ODEs, the initial condition $\mathbf{x}(0)$ doesn't uniquely determine the trajectory.
- There are two sources of randomness:
  - the initial distribution $p_0(\mathbf{x})$;
  - the Wiener process $\mathbf{w}(t)$.

# Stochastic Differential Equation (SDE)

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)\, dt + g(t)\, d\mathbf{w}$$

## Discretizing the SDE (Euler-Maruyama Update) – `SDESolve`

$$\mathbf{x}(t + dt) = \mathbf{x}(t) + \mathbf{f}(\mathbf{x}(t), t) \cdot dt + g(t) \cdot \boldsymbol{\epsilon} \cdot \sqrt{dt}$$

If $dt = 1$, then

$$\mathbf{x}_{t+1} = \mathbf{x}_t + \mathbf{f}(\mathbf{x}_t, t) + g(t) \cdot \boldsymbol{\epsilon}$$

- At any time $t$, the process has density $p_t(\mathbf{x}) = p(\mathbf{x}, t)$.
- $p : \mathbb{R}^m \times [0, 1] \to \mathbb{R}_+$ specifies a **probability path** from $p_0(\mathbf{x})$ to $p_1(\mathbf{x})$.
- How can we obtain the probability path $p_t(\mathbf{x})$ for $\mathbf{x}(t)$?

# Stochastic Differential Equation (SDE)

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)\, dt + g(t)\, d\mathbf{w}, \quad d\mathbf{w} = \boldsymbol{\epsilon} \cdot \sqrt{dt}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}).$$

**Theorem (Kolmogorov-Fokker-Planck)**

If $p_t(\mathbf{x}) \in C^{1,2}$ (i.e., $C^1$ in $t$ and $C^2$ in $\mathbf{x}$), then

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}\left(\mathbf{f}(\mathbf{x}, t) p_t(\mathbf{x})\right) + \frac{1}{2} g^2(t) \Delta_{\mathbf{x}} p_t(\mathbf{x})$$

Here,

$$\operatorname{div}(\mathbf{v}) = \sum_{i=1}^{m} \frac{\partial v_i(\mathbf{x})}{\partial x_i} = \operatorname{tr}\left(\frac{\partial \mathbf{v}(\mathbf{x})}{\partial \mathbf{x}}\right)$$

$$\Delta_{\mathbf{x}} p_t(\mathbf{x}) = \sum_{i=1}^{m} \frac{\partial^2 p_t(\mathbf{x})}{\partial x_i^2} = \operatorname{tr}\left(\frac{\partial^2 p_t(\mathbf{x})}{\partial \mathbf{x}^2}\right)$$

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = \operatorname{tr}\left(-\frac{\partial}{\partial \mathbf{x}}\left[\mathbf{f}(\mathbf{x}, t) p_t(\mathbf{x})\right] + \frac{1}{2} g^2(t) \frac{\partial^2 p_t(\mathbf{x})}{\partial \mathbf{x}^2}\right)$$

# Stochastic Differential Equation (SDE)

## Theorem (Kolmogorov-Fokker-Planck)

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = \operatorname{tr}\left(-\frac{\partial}{\partial \mathbf{x}}\left[\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})\right] + \frac{1}{2}g^2(t)\frac{\partial^2 p_t(\mathbf{x})}{\partial \mathbf{x}^2}\right)$$

- The KFP theorem is a necessary and sufficient condition (it uniquely defines $p_t(\mathbf{x})$).
- This generalizes the continuity equation for continuous-time NF:

$$\frac{d \log p_t(\mathbf{x}(t))}{dt} = -\operatorname{tr}\left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right).$$

## Special Case: constant density $p_t(\mathbf{x})$

Let's find the SDE for which $p_t(\mathbf{x}) = \text{const}$ (i.e., if $\mathbf{x}(0) \sim p_0(\mathbf{x})$, then $\mathbf{x}(t) \sim p_0(\mathbf{x})$.).

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = 0 \quad \Leftrightarrow \quad \operatorname{tr}\left(-\frac{\partial}{\partial \mathbf{x}}\left[\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})\right] + \frac{1}{2}g^2(t)\frac{\partial^2 p_t(\mathbf{x})}{\partial \mathbf{x}^2}\right) = 0$$

# Langevin SDE (Special Case)

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = 0 \quad \Leftrightarrow \quad \operatorname{tr}\left(-\frac{\partial}{\partial \mathbf{x}}\left[\mathbf{f}(\mathbf{x}, t) p_t(\mathbf{x})\right] + \frac{1}{2} g^2(t) \frac{\partial^2 p_t(\mathbf{x})}{\partial \mathbf{x}^2}\right) = 0$$

$$\frac{\partial}{\partial \mathbf{x}}\left[\mathbf{f}(\mathbf{x}, t) p_t(\mathbf{x})\right] = \frac{1}{2} g^2(t) \frac{\partial^2 p_t(\mathbf{x})}{\partial \mathbf{x}^2}$$

$$\mathbf{f}(\mathbf{x}, t) p_t(\mathbf{x}) = \frac{1}{2} g^2(t) \frac{\partial p_t(\mathbf{x})}{\partial \mathbf{x}}$$

$$\mathbf{f}(\mathbf{x}, t) = \frac{1}{2} g^2(t) \frac{1}{p_t(\mathbf{x})} \frac{\partial p_t(\mathbf{x})}{\partial \mathbf{x}} = \frac{1}{2} g^2(t) \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})$$

Let $g(t) = 1$, then $\mathbf{f}(\mathbf{x}, t) = \frac{1}{2} \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})$.

$$d\mathbf{x} = \frac{1}{2} \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})\, dt + 1 \cdot d\mathbf{w}$$

# Langevin SDE (Special Case)

Let's find the SDE for which $p_t(\mathbf{x}) = \text{const}$ (i.e., if $\mathbf{x}(0) \sim p_0(\mathbf{x})$, then $\mathbf{x}(t) \sim p_0(\mathbf{x})$.).

$$d\mathbf{x} = \frac{1}{2}\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})\, dt + 1 \cdot d\mathbf{w}$$

## Discretized Langevin SDE

$$\mathbf{x}_{t+1} - \mathbf{x}_t = \frac{\eta}{2} \cdot \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}, \quad \eta \approx dt.$$

Setting the stationary density to the model distribution, $p_t(\mathbf{x}) = p_{\boldsymbol{\theta}}(\mathbf{x})$ for all $t$:

## Langevin Dynamic

$$\mathbf{x}_{t+1} = \mathbf{x}_t + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}} \log p_{\boldsymbol{\theta}}(\mathbf{x}) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}, \quad \eta \approx dt.$$

We (partially) explained, why Langevin dynamics is working.

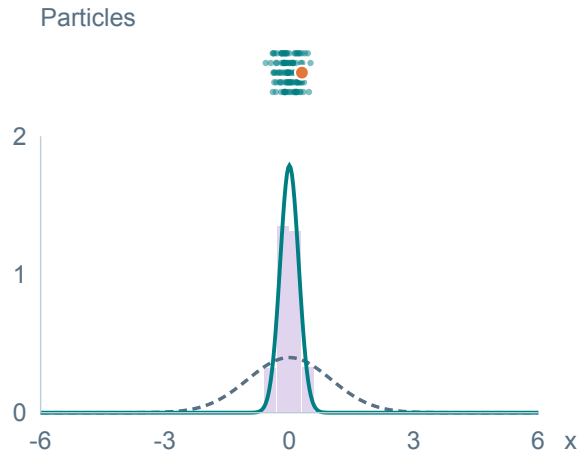
# Langevin: Moving Particles, Stationary Density

$$p_0(x) = p_*(x) = \mathcal{N}(0, 1), \qquad \tfrac{1}{2}\nabla_x \log p_*(x) = -\tfrac{1}{2}x$$

Same initial samples from the target. All three cases at t = 3.

Empirical histogram — Exact current density --- Target N(0, 1)

**Drift only** · $dx = -\frac{1}{2}x\,dt$



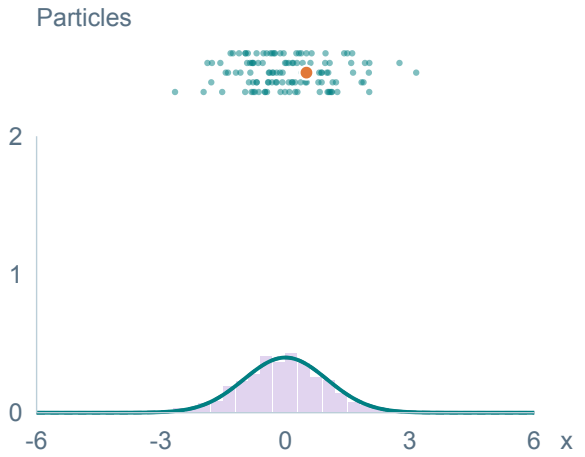

Exact variance: 0.050

**Diffusion only** · $dx = dw$




Exact variance: 4.000

**Both** · $dx = -\frac{1}{2}x\,dt + dw$




Exact variance: 1.000

> Stationary density does not mean stationary particles.

768 samples; finite-sample fluctuations remain. Exact Gaussian transitions isolate stationarity from discretization error.

# Summary

- CNFs define invertible transformations via neural ODEs; likelihood-based training requires ODE solves, trace evaluations and gradients through the solver.

- If $\mathbf{x}_0$ is a random variable, this yields a **probability path** $p_t(\mathbf{x})$ as time evolves. The continuity equation describes the evolution of $\log p(\mathbf{x}, t)$ over time.

- An SDE defines a stochastic process with drift and diffusion terms; ODEs are a special case of SDEs.

- The KFP equation describes the probability dynamics of an SDE.

- The Langevin SDE preserves a constant probability path.